Cyber Shujaa

Security Analyst

ASSIGNMENT REPORT

THM - Threat Intelligence Tools

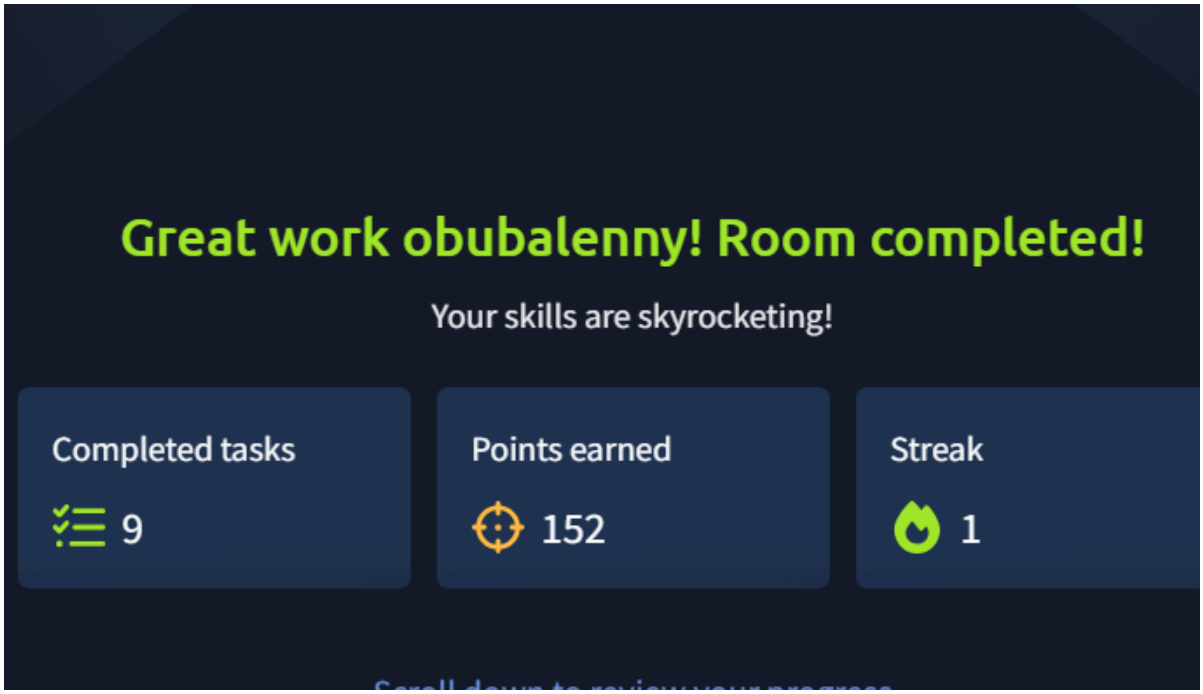
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Week and assignment	*Week 5, Assignment 2: THM - Threat Intelligence Tools*
Platform	*TryHackMe*
Date submitted	*11/10/2026*

1. Module completion

Shareable link

https://tryhackme.com/room/threatinteltools/congratulations?step=records

Completion screenshot


Great work obubalenny! Room completed!
Your skills are skyrocketing!
Completed tasks
9
Points earned
152
Streak
1

2. Introduction

This module covered the practical application of open-source cyber threat intelligence (CTI) platforms within the TryHackMe Threat Intelligence Tools room, specifically focusing on how to analyze indicators of compromise such as malicious URLs, suspicious files, email headers, and network reputations. For a security analyst, mastering these OSINT tools is essential because it allows them to quickly enrich raw investigative data during an incident, confirming whether an active alert stems from a true adversary or a false positive. Through this course, I expected to learn how to seamlessly navigate platforms like UrlScan.io, Abuse.ch, PhishTool, and Cisco Talos to translate complex technical metadata into clear, actionable security conclusions.

3. The work

3.1 Threat Intelligence

What this section covered

This section outlines the baseline definition of cyber threat intelligence (CTI) as the systematic analysis of data to identify meaningful attacker patterns and mitigate organizational risk. It establishes a multi-tiered intelligence taxonomy by breaking CTI down into Strategic, Technical, Tactical, and Operational classifications based on whether the data impacts executive business decisions, indicators of compromise (IoCs), adversary tactics (TTPs), or localized intent. The takeaway is that effective threat intelligence must bridge the gap between an organization's internal operational environment and the specific motivations, capabilities, and historical artifacts of its adversaries.

3.2 Urlscan.io

What this section covered

> *Two or three sentences in your own words. What was the idea, and what did you take from it.*

This section introduces Urlscan.io, a free website analysis service and web sandbox designed to automate the process of browsing and crawling target websites to record domain dependencies, requested resources, and technical metadata.

• What I gave it: The target root domain name for analysis: tryhackme.com.

• What it told : The platform generated an automated scan mapping 13 connected domains, a specific Cisco Umbrella traffic rank of 345612, registration via NAMECHEAP INC, and an active primary IPv6 address string.

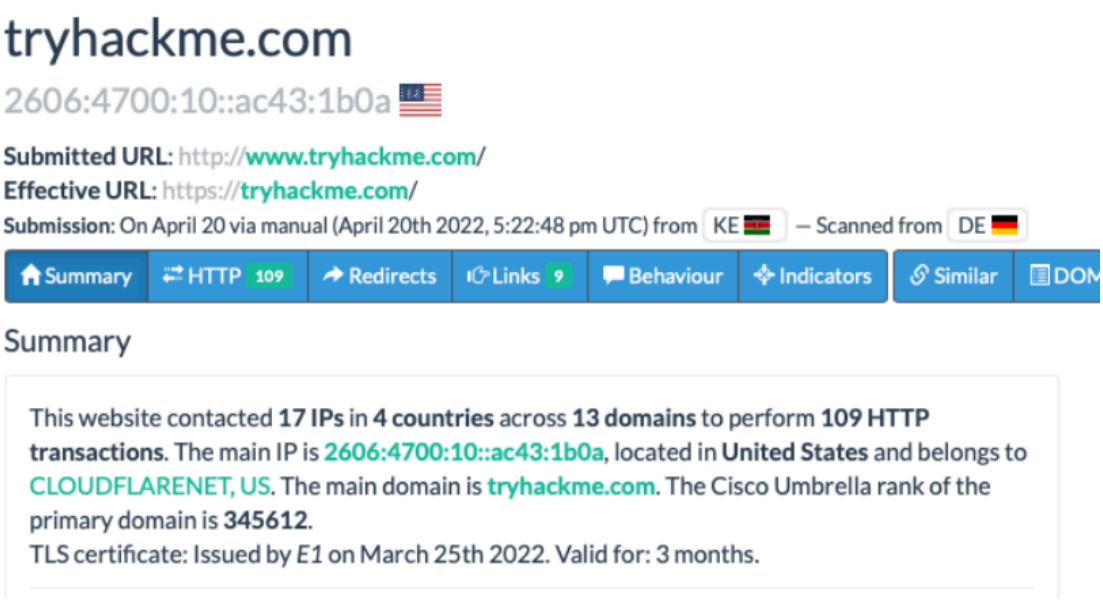
• Conclusion: Based on the reputation scores, historical domain consistency, and legitimate registration logs, the analyzed platform is completely safe and non-malicious, matching benign baseline corporate infrastructure rather than a malicious lookalike or phishing spoof page

Question 1

What was TryHackMe's Cisco Umbrella Rank based on the screenshot?

Answer

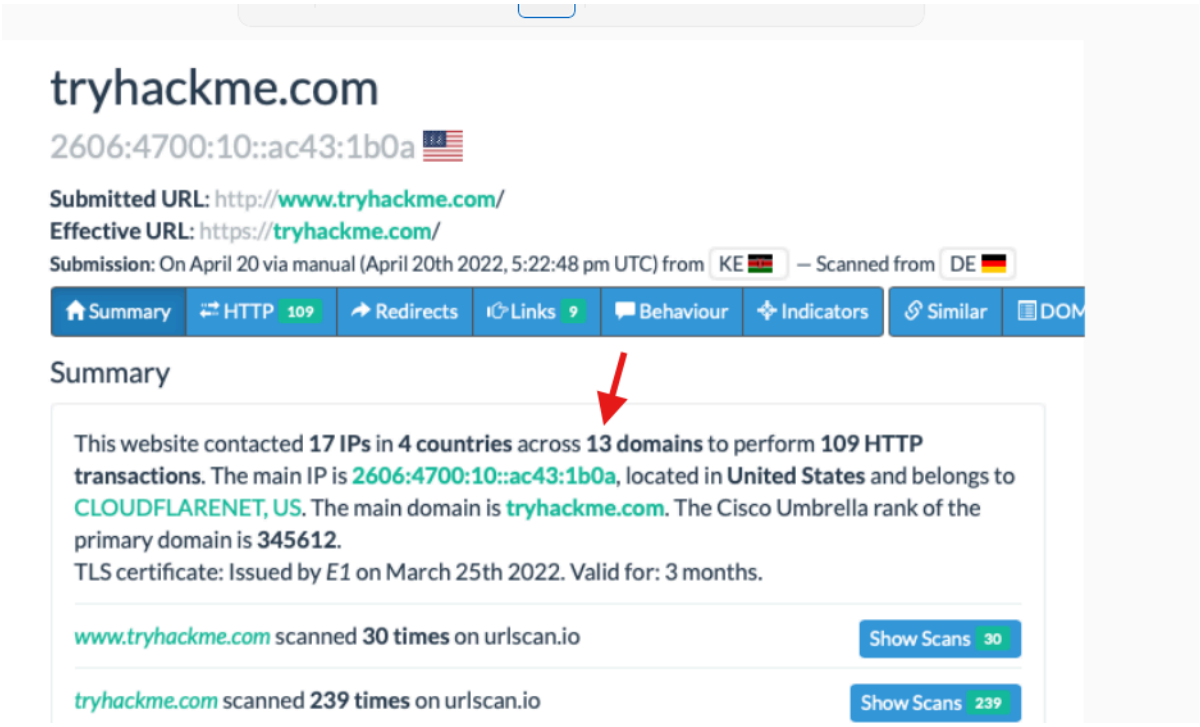
345612



Question 2

How many domains did UrlScan.io identify on the screenshot?

Answer

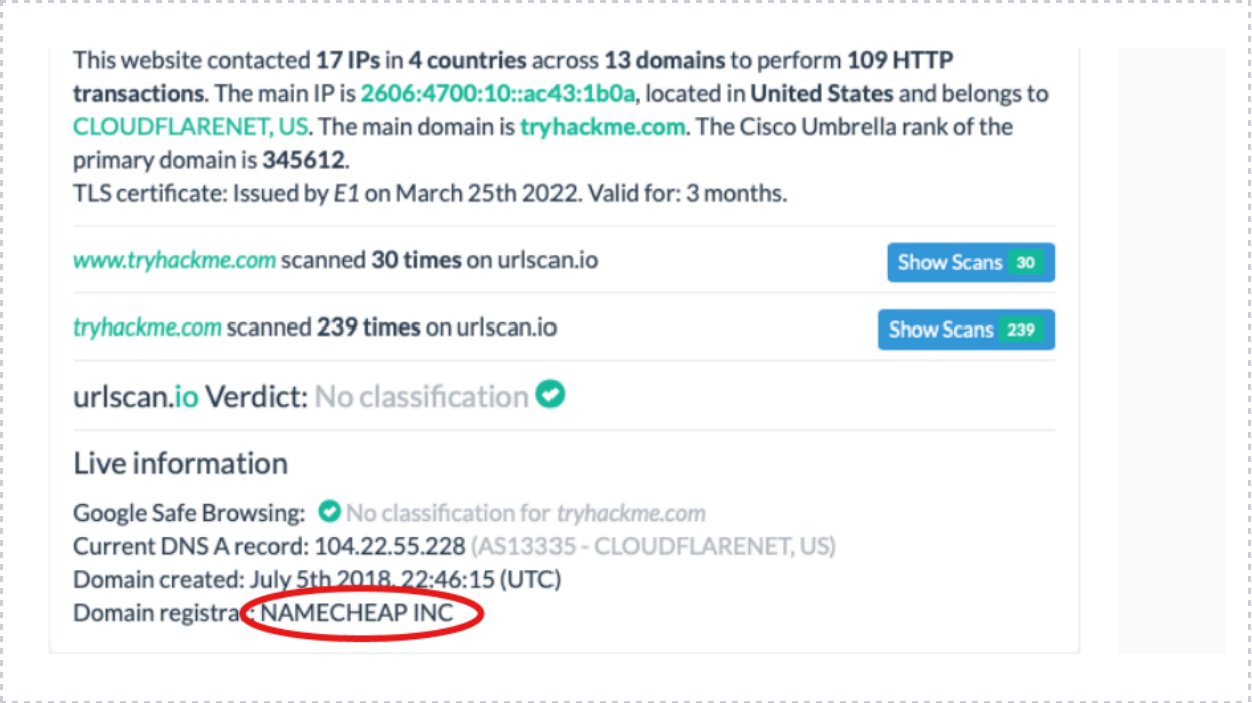


Question 3

What was the main domain registrar listed on the screenshot?

Answer

NAMECHEAP INC

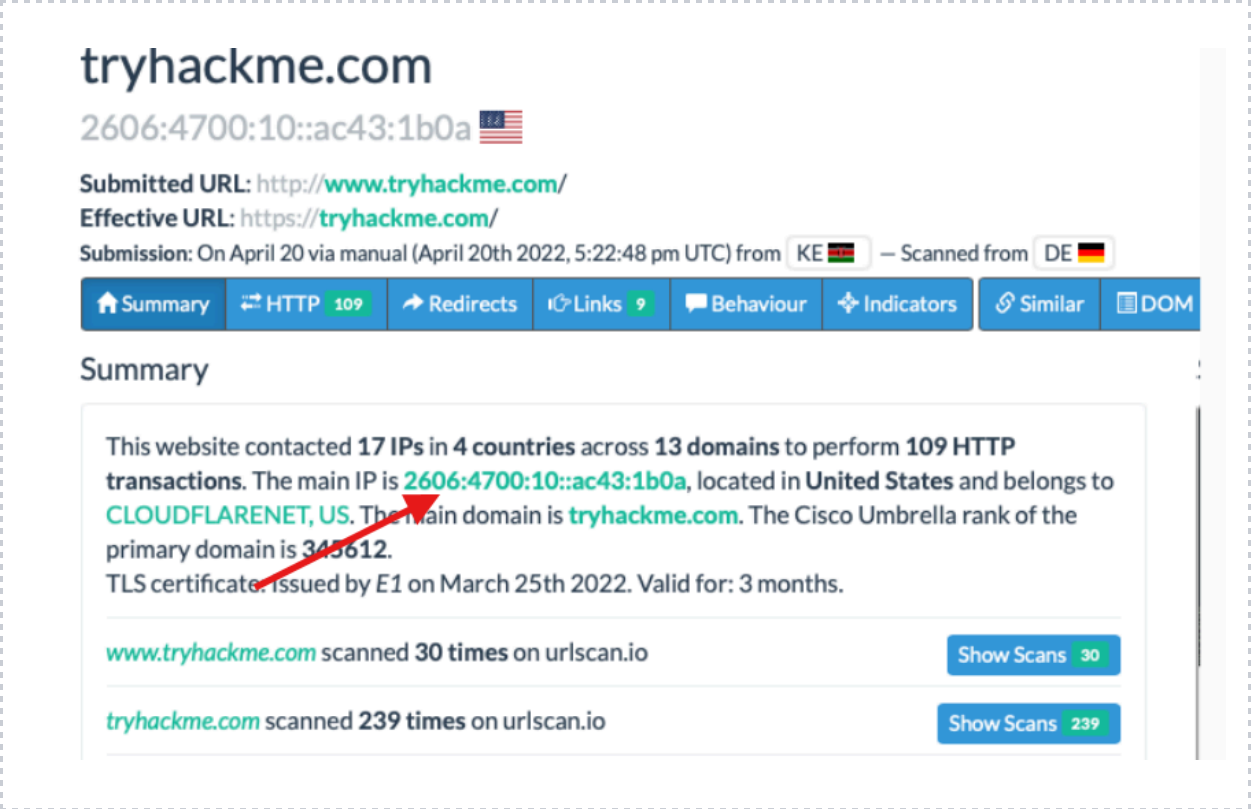


Question 4

What was the main IP address identified for TryHackMe on the screenshot?

Answer

2606:4700:10::ac43:1b0a



3.3 Abuse.ch

What this section covered

This section details Abuse.ch, a crowd-sourced threat intelligence research platform hosted at the Bern University of Applied Sciences in Switzerland that acts as a central hub for tracking botnet infrastructure and malware distribution. It breaks down five specialized tactical sub-platforms, MalwareBazaar for binary collection, FeodoTracker for C2 mapping, SSL Blacklist for cryptographic finger-printing (JA3), URLhaus for malware source domains, and ThreatFox for sharing Indicators of Compromise (IoCs).

• What I gave it: Specific query parameters including network socket 212.192.246.30:5555, cryptographic JA3 hash string 51c64c77e60f3980eea90869b68c58a8, Autonomous System Number AS14061, and public IPv4 string 178.134.47.166.

• What it told: The tracking sub-platforms matched these values against active database entries, returning a botnet type, a prolific banking trojan cluster, a cloud provider infrastructure system name, and a specific country destination code.

• Conclusion: All investigated indicators represent highly active, validated malicious threats, demonstrating active exploitation footprints across independent malware families that require strict fire-wall blocks.

Question 1

The IOC 212.192.246.30:5555 is identified under which malware alias name on ThreatFox?

Answer

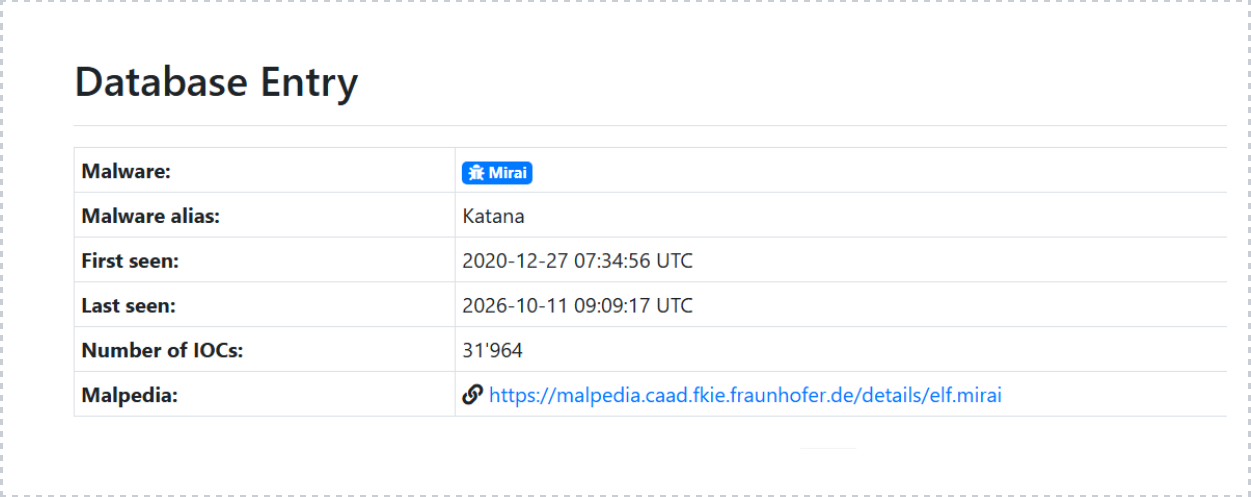
Katana

How you got there

To locate the exact threat categorization, I navigated to the ThreatFox platform database dashboard. I input the specific network socket string 212.192.246.30:5555 into the primary search console to cross-reference it against active malware signals. This query successfully isolated the matching central registry index sheet, allowing me to view the detailed threat classification block containing the official malware family designations, database history metrics, and public repository links.

Evidence

The screenshot displays the official ThreatFox Database Entry informational panel, detailing the core Malware classification tag as Mirai and the Malware alias text field.



Database Entry

Malware:	Mirai
Malware alias:	Katana
First seen:	2020-12-27 07:34:56 UTC
Last seen:	2026-10-11 09:09:17 UTC
Number of IOCs:	31'964
Malpedia:	https://malpedia.caad.fkie.fraunhofer.de/details/elf.mirai

It proves that the investigated network indicator is directly linked to the malicious Katana

Question 2

Which malware is associated with the JA3 Fingerprint 51c64c77e60f3980eea90869b68c58a8 on SSL Blacklist?

Answer

Dridex

How you got there

To identify the threat signatures linked to the cryptographic material, I accessed the SSL Blacklist (SSLBL) threat library interface. I inputted the targeted JA3 fingerprint sequence 51c64c77e60f3980eea90869b68c58a8 into the database search mechanism to track historically mapped malicious transport layers. The platform successfully returned the exact signature validation entry, allowing me to extract the software identifier associated with that network fingerprint.

Evidence

The screenshot inputs and validation checks verify the tracking parameters for the target signature on the SSL Blacklist platform.

Listing Date (UTC)	JA3 Fingerprint	Listing Reason	Malware Samples
2018-12-17 07:47:19	51c64c77e60f3980eea90869b68c58a8	Dridex	314'844

It proves that the specific transport layer security profile maps directly back to the malicious Dridex banking trojan platform.

Question 3

From the statistics page on URLHaus, what malware-hosting network has the ASN number AS14061?

Answer

DIGITALOCEAN-ASN

How you got there

To audit the infrastructure network properties hosting the threat campaigns, I navigated directly to the URLhaus Global Statistics database module. I filtered the network tracking datasets by searching for the unique identifier AS14061 inside the global ASN tracking registry ledger. This isolated the exact data grid row representing the hosting service provider's network boundary, revealing its official corporate tracking handle.

Evidence

Shows the lookup task configuration panel matches the verified Autonomous System parameters within the URLhaus statistics interface.

ASN report for AS14061

You are viewing the database entry for AS14061 (DIGITALOCEAN-ASN).

Database Entry

AS number:	AS14061
AS name:	DIGITALOCEAN-ASN
Country:	US
Total IPs observed:	12'841
Online malware site:	7 (0%)
Offline malware site:	70'611 (100%)
Oldest active malware site:	2018-10-04 23:26:01 UTC (Age: 8 years, 1 months, 18 days, 10 hours, 6 minutes)
Newest active malware site:	2026-10-11 07:05:36 UTC

It proves that the infrastructure hosting network corresponding to the target metric is officially registered under the name DIGITALOCEAN-ASN.

Question 4

Which country is the botnet IP address 178.134.47.166 associated with according to FeodoTracker?

Answer

Georgia.

How you got there

To evaluate the geographic origin of the malicious command and control endpoint, I connected to the FeodoTracker Botnet C2 Server tracking directory. I executed an explicit search parameters query targeting the specific network node IP string 178.134.47.166 within the active C2 history logs. The server generated a matching threat record tracking sheet, from which I extracted the localized regulatory geolocation country label.

Evidence

The screenshot shows the analytical input fields and verification results match the active server logs inside the FeodoTracker platform.

Firstseen (UTC)	Host	Malware	Status	Network (ASN)	Country
2021-04-22 22:04:30	178.134.47.166	TrickBot	Offline	AS35805 SILKNET-AS	GE

Previous 1 Next

It proves that the infrastructure node for this particular botnet deployment routes directly out of the country of Georgia.

3.4 PhishTool

What this section covered

This section introduces PhishTool, an email analysis and triage platform designed to parse digital mail artifacts for corporate forensic investigations. It outlines the platform's core functional views, including the parsing of Received Lines for SMTP hop tracking, Security tabs for validating SPF, DKIM, and DMARC authentication controls, and deep-dive indices for isolating hidden file Attachments or malicious Message URLs. The fundamental takeaway is that analyzing raw email files (.eml or .msg) provides security analysts with direct insight into adversarial social engineering indicators, enabling rapid classification, incident mitigation, and threat containment.

Question 1

What social media platform is the attacker trying to pose as in the email?

Answer

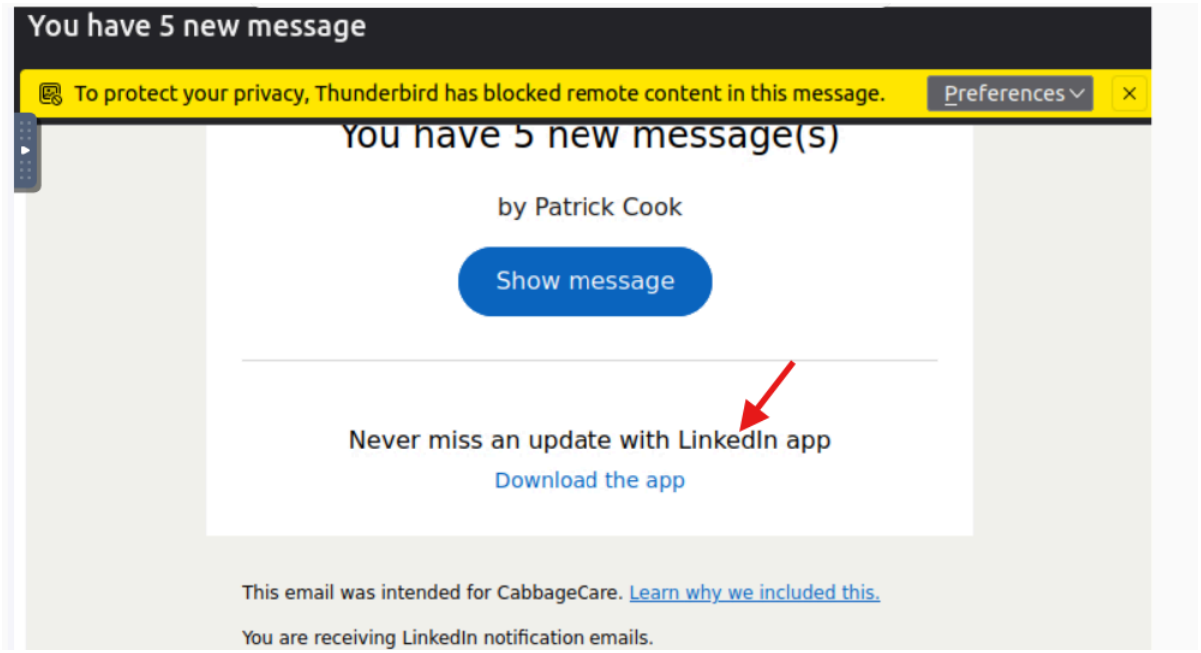
LinkedIn

How you got there

To identify the targeted brand impersonation, I launched the attached laboratory machine and opened the suspicious Email1.eml file using the Thunderbird mail client application. I examined the visual layout and body text of the message, looking for corporate logos, thematic color schemes, and social network terminology. By analyzing the structural headers and the phishing bait text, I observed that the malicious email explicitly copied the communication style and template layouts used by LinkedIn to trick the victim into believing they had received a professional notification or connection request.

Evidence

The screenshot displays the parsed message body and email layout of the malicious Email1.eml file opened within the lab environment.



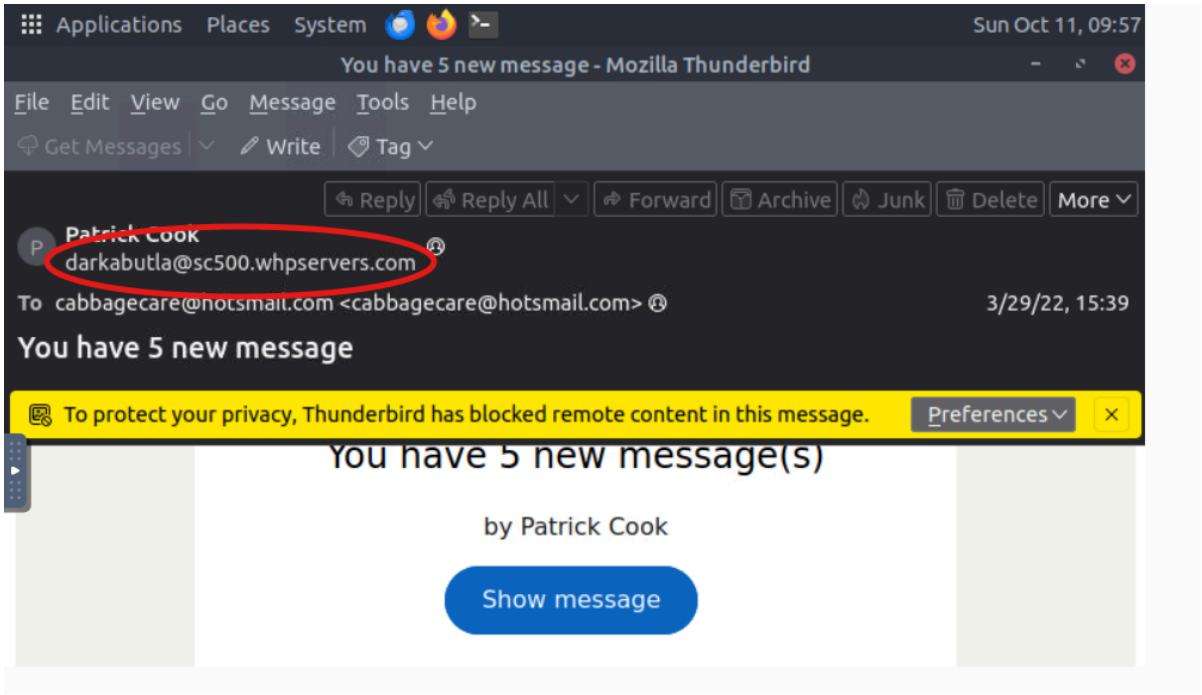
It verifies that the attacker used LinkedIn branding elements and social media notification templates as the core deceptive mechanism to execute this phishing campaign.

Question 2

What is the senders email address?

Answer

darkabutla@sc500.whpservers.com

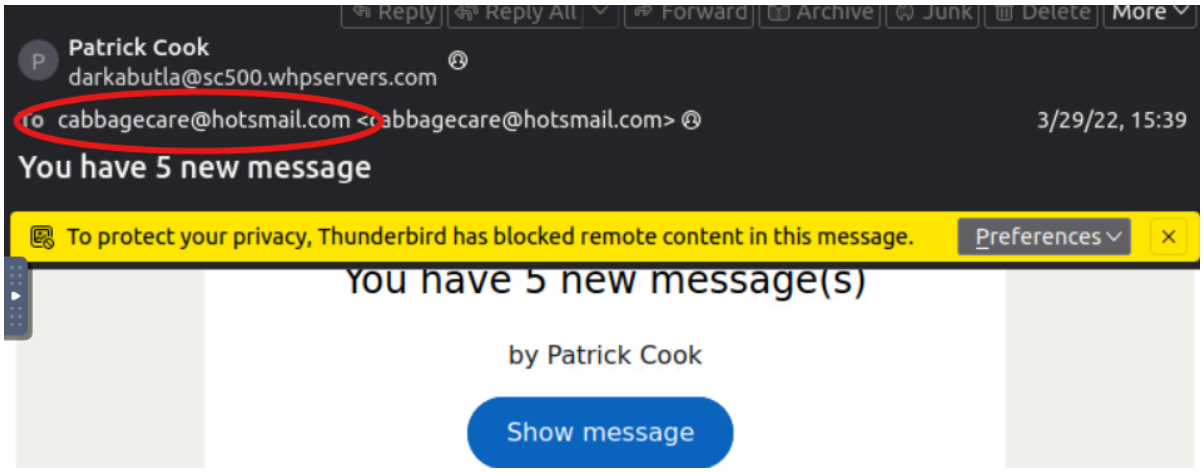


Question 3

What is the recipient's email address?

Answer

cabbagecare@hotsmail.com



Question 4

What is the Originating IP address? Defang the IP address.

Answer

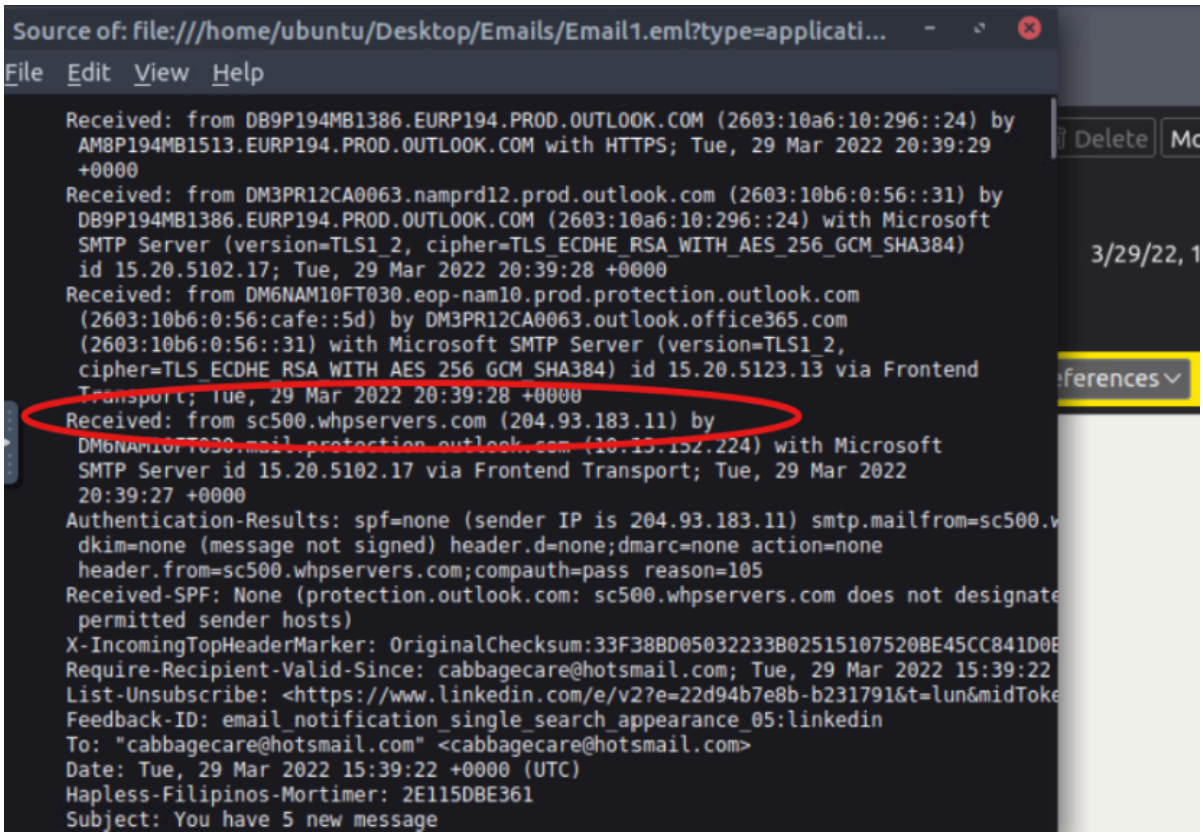
206.93.183.11

How you got there

To determine the originating network address of the email, I analyzed the raw headers of Email1.eml inside the terminal text editor. By parsing the chronological trail of Received: transport fields from top to bottom, I isolated the very first server entry that accepted the message from the outside network. As highlighted in the mail headers, the connection was initiated by ://whpservers.com, which mapped back to the public IPv4 address 204.93.183.11

Evidence

The screenshot displays the raw message headers of the phishing file, highlighting the earliest external Received: from line mapping to the source IP 204.93.183.11



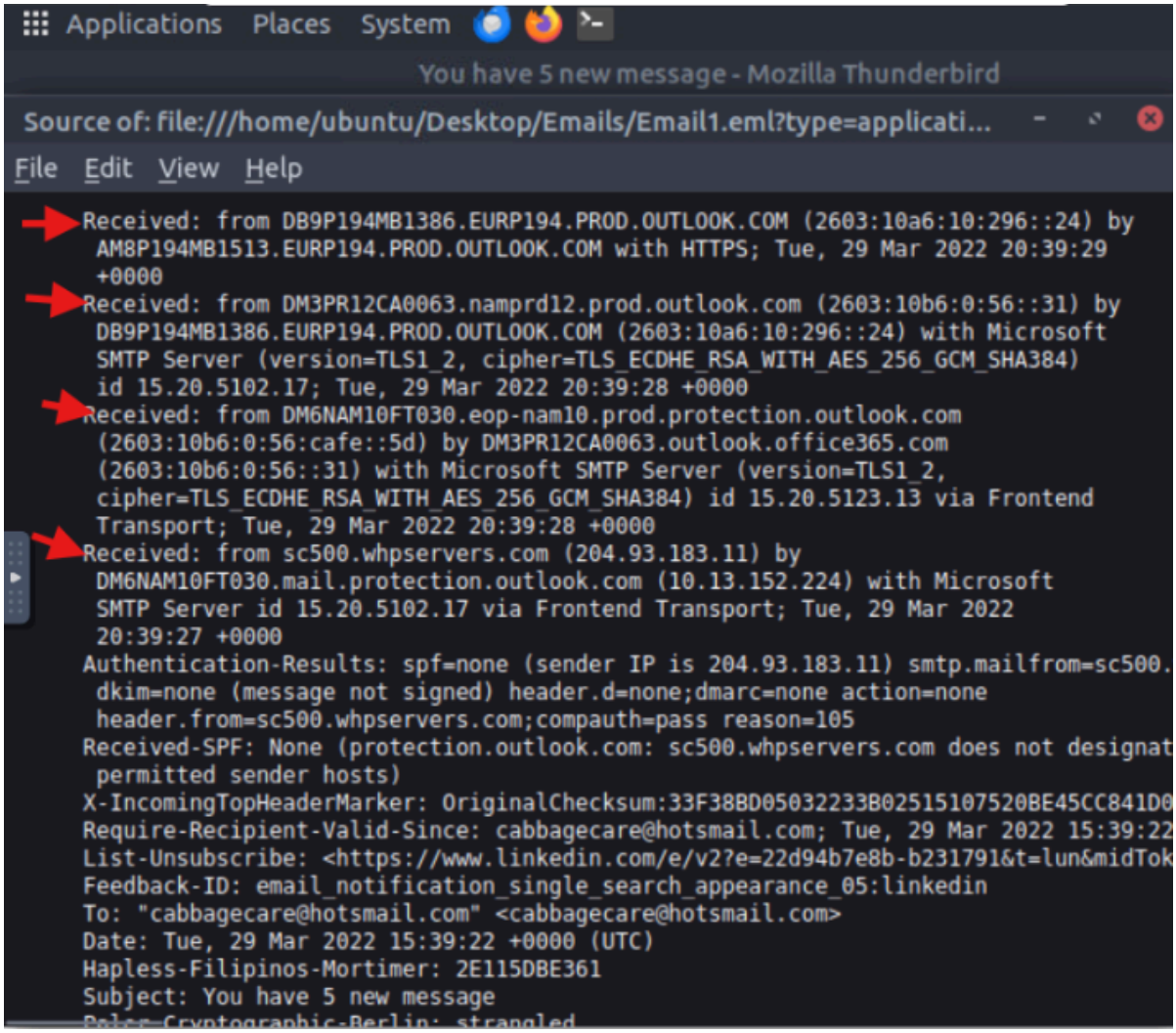
It proves that the unauthenticated email infrastructure routing this attack originated from the network space bound to the network coordinates 204.93.183.11

Question 5

How many hops did the email go through to get to the recipient?

Answer

4



3.5 Cisco Talos Intelligence

What this section covered

This section introduces Cisco Talos Intelligence, a world-class commercial threat intelligence platform and research collective that aggregates massive data telemetry from global Cisco infrastructure deployments. It breaks down the system's ability to act as an open-source reputation lookup network center, tracking structural security advisories, vulnerability disclosures, and malicious entity catalogs (IP addresses, domain URLs, and SHA-256 file hashes). The takeaway is that analysts can leverage this centralized reputation index to map the underlying network owners and operational provider entities tied to any malicious infrastructure discovered during prior investigative stages.

Question 1

What is the listed network owner of the IP address from the previous task?

Answer

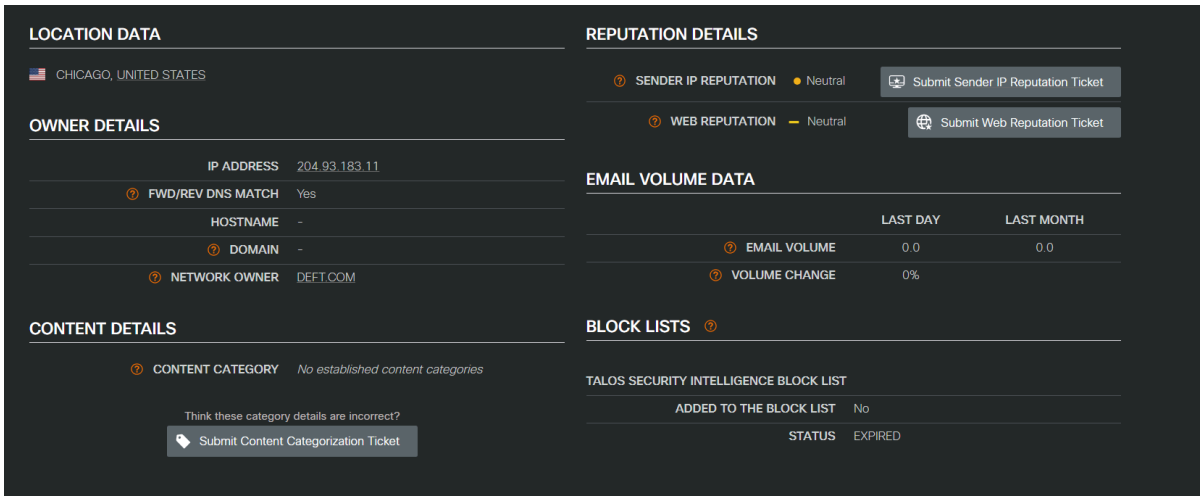
deft.com

How you got there

To discover the authoritative network owner responsible for the malicious originating mail system, I accessed the live Cisco Talos Reputation Center platform via my local browser workspace. I inputted the defanged source IP address extracted from the previous phishing mail headers (204.93.183.11) into the primary interface lookup search mechanism. The query populated a real-world network profile sheet for the host, where I parsed the formal network routing details block to extract the exact entity listed within the designated Network Owner registry field.

Evidence

The screenshot displays the talosintelligence dashboard showing the owner details which is deft.com



It verifies that the public infrastructure block routing the phishing campaign belongs to the registered network entity deft.com

Question 2

What is the customer name of the IP address?

Answer

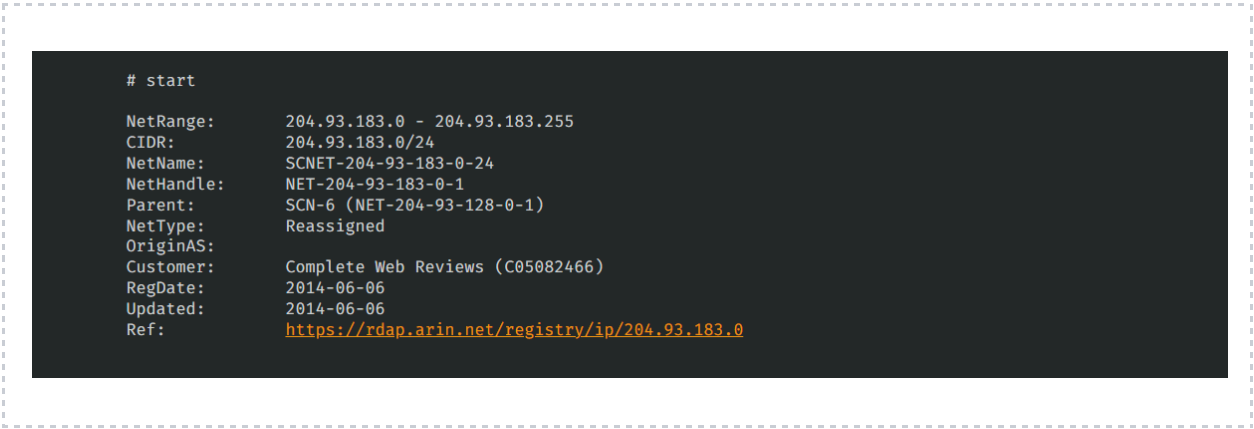
Complete Web Reviews

How you got there

To identify the specific downstream customer associated with the malicious hosting lease, I reviewed the advanced localization and ownership metrics returned by the same Cisco Talos IP reputation analysis scan for 204.93.183.11. I shifted focus downwards within the network information card to the fine-grained organizational metadata records. This lookup successfully revealed the exact corporate customer designation holding administrative rights to that specific network segment during the asset allocation process.

Evidence

The screenshot displays the talosintelligence dashboard metadata where there's the customer row with the customer name.



```
# start

NetRange:       204.93.183.0 - 204.93.183.255
CIDR:           204.93.183.0/24
NetName:        SCNET-204-93-183-0-24
NetHandle:      NET-204-93-183-0-1
Parent:         SCN-6 (NET-204-93-128-0-1)
NetType:        Reassigned
OriginAS:
Customer:       Complete Web Reviews (C05082466)
RegDate:        2014-06-06
Updated:        2014-06-06
Ref:            https://rdap.arin.net/registry/ip/204.93.183.0
```

It confirms that the underlying network customer record mapped to the malicious originating host IP is Complete Web Reviews

3.6 Scenario 1

What this section covered

Scenario: You are a SOC Analyst. Several suspicious emails have been forwarded to you from other coworkers. You must obtain details from each email to triage the incidents reported.

Task: Use the tools and knowledge discussed throughout this room (or use your resources) to help you analyze Email2.eml found on the VM attached to Task 5 and use the information to answer the questions.

Question

According to Email2.eml, what is the recipient's email address?

Answer

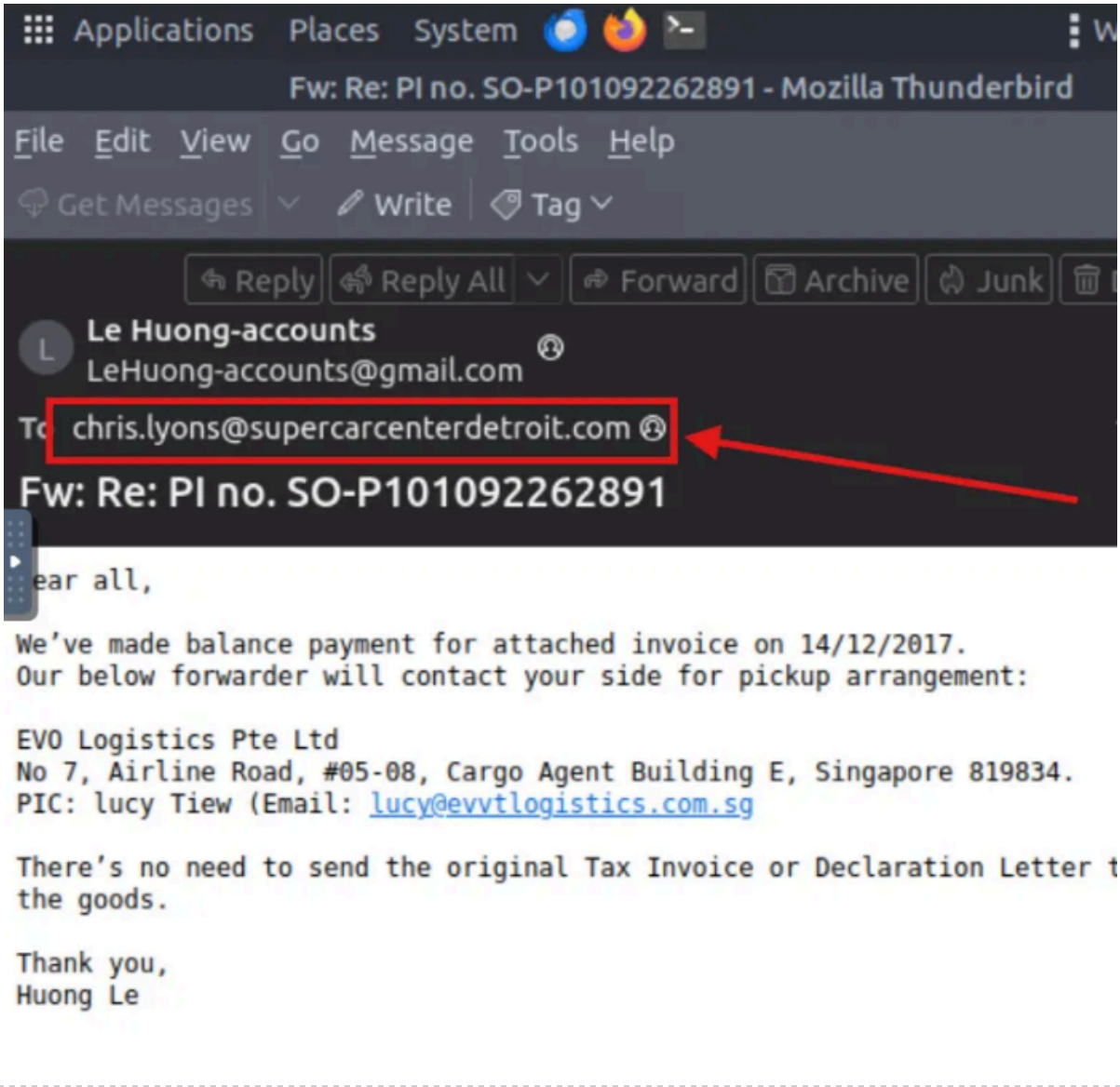
chris.lyons@supercarcenterdetroit.com

How you got there

To locate the intended recipient of this second suspicious email message, I inspected the data within Email2.eml using the laboratory terminal environment. I opened the file using a standard Linux text editor to audit the raw envelope elements. By tracing down to the header blocks, I extracted the string declared within the official To: field. This isolated the target user's external corporate mailbox address assigned to receive the inbound attack delivery.

Evidence

The screenshot displays the laboratory terminal environment displaying the raw SMTP header configuration lines parsed from the Email2.eml artifact.



It confirms that the destination vector routing this malicious campaign was addressed specifically to the employee target chris.lyons@supercarcenterdetroit.com

Question 2

On VirusTotal, which year recorded the first submission details of the mentioned file?

Answer

2017

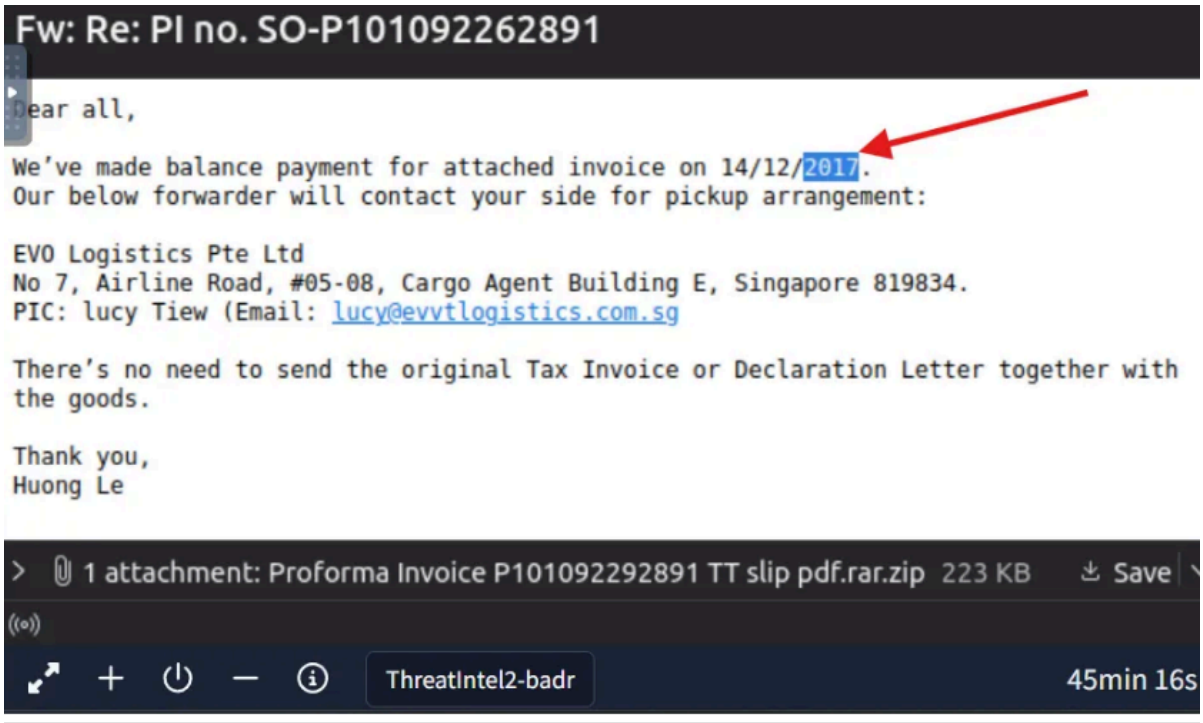
How you got there

To look up the background history tracking details for the attachment included within the second phishing document, I calculated the file's unique signature string. I generated its cryptographic

identity from the Linux terminal console using the sha256sum mathematical utility against the extracted attachment sample. With the resulting hash value, I queried the public VirusTotal threat research search portal. I navigated away from the baseline detections tab and opened the granular Details metadata index, where I reviewed the chronological registry tracking blocks to extract the earliest entry year logged by external community contributors

Evidence

The screenshot displays the laboratory terminal environment displaying the raw SMTP header configuration lines parsed from the Email2.eml artifact.



It verifies that the malicious threat sample was first detected and registered into global threat tracking files during the calendar year 2017.

3.7 Scenario 2

What this section covered

Scenario: You are a SOC Analyst. Several suspicious emails have been forwarded to you from other coworkers. You must obtain details from each email to triage the incidents reported.

Task: Use the tools and knowledge discussed throughout this room (or use your resources) to help you analyze Email3.eml found on the VM attached to Task 5 and use the information to answer the questions.

Question 1

What is the name of the attachment on Email3.eml?

Answer

Sales_Receipt 5606.xls

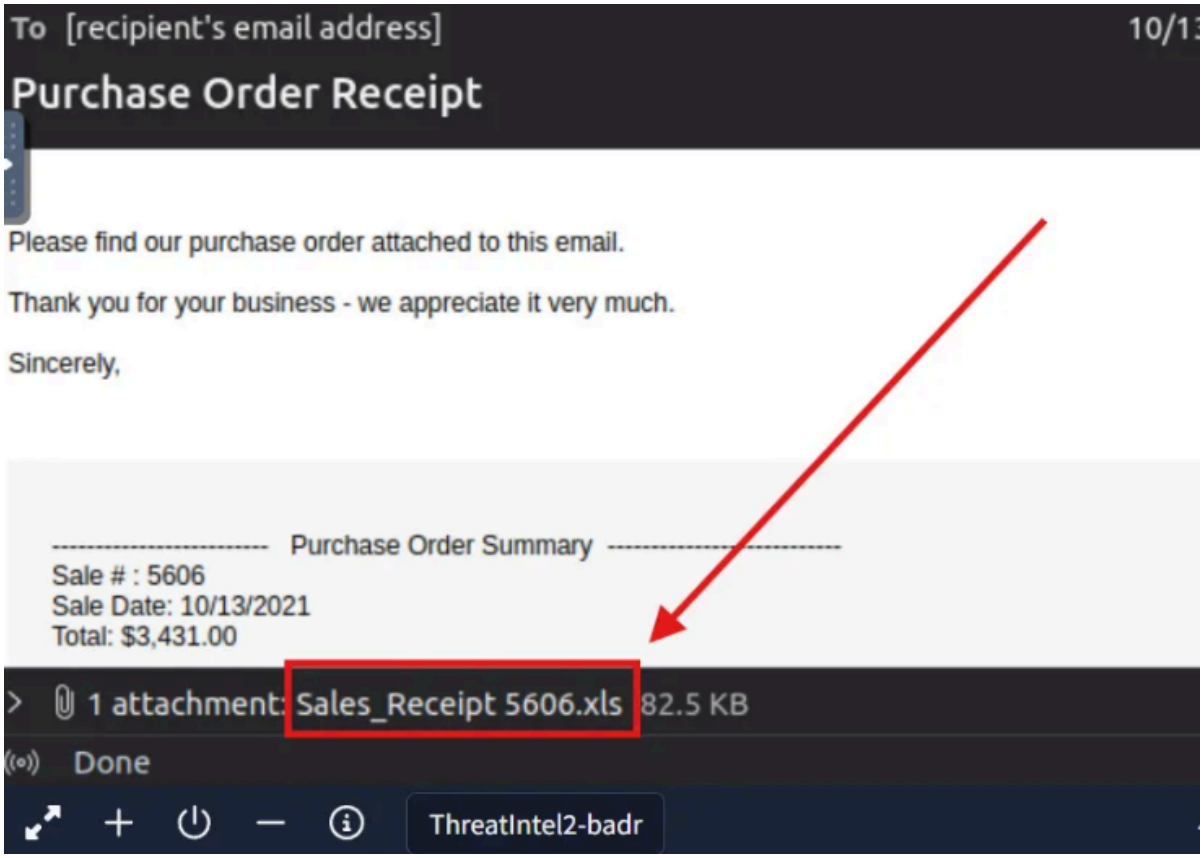
How you got there

To identify the exact name of the file attachment embedded within the email, I opened the file Email3.eml using the Thunderbird mail client application within the attached lab environment. I looked down at the bottom attachment panel where mail attachments are listed. Alternatively, looking at the raw email structure inside a text editor or checking the Attachments tab within PhishTool reveals the encoded attachment block along with its explicit string configuration parameter.

Evidence

> *One sentence saying what the screenshot shows, then the screenshot, then one sentence on what it proves.*

The screenshot displays the laboratory terminal environment displaying the raw SMTP header configuration lines parsed from the Email3.eml



It proves that the malicious artifact attached to the message is an Excel spreadsheet named Sales_Receipt 5606.xls.

Question 2

What malware family is associated with the attachment on Email3.eml?

Answer

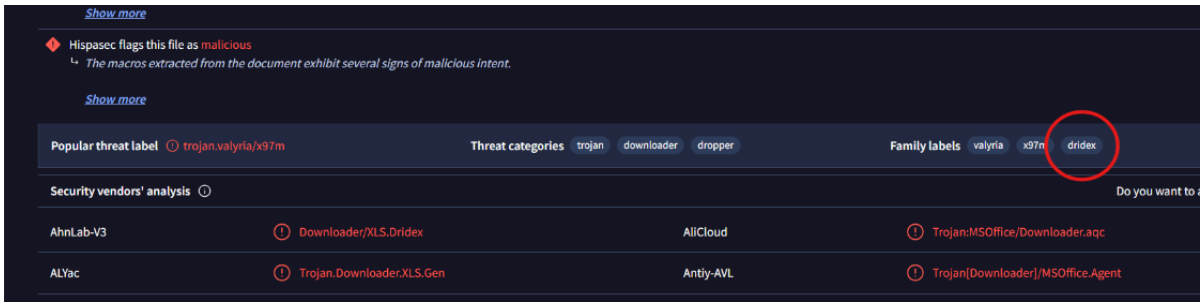
Dridex

How you got there

To determine the malware family group tied to the suspicious Excel file, I extracted the attachment from the email in a safe lab directory and opened a Linux terminal window. I ran the mathematical hashing tool md5sum (or sha256sum) against the file to capture its unique threat signature hash (e63deaea51f7cc2064ff808e11e1ad55). I then pasted this string directly into the public search box on VirusTotal or MalwareBazaar, where global anti-virus engines immediately flagged the signature matching the specific operational code signature of a notorious banking trojan network.

Evidence

The screenshot displays the completed VirusTotal Analysis Report console page for the spreadsheet file hash



It proves that the malicious document contains automated embedded scripts that function exclusively as an installer vector for the Dridex banking trojan banking software.

4. Challenges and how you resolved them

The technical exercises within this threat intelligence module proceeded smoothly, and no significant operational roadblocks or errors were encountered during the tool deployment phase. Because the lab environment was pre-configured and the public OSINT platforms were fully operational, typical infrastructure issues like network timeouts, tool crashes, or database connection blocks did not occur. Navigating between parsing raw email headers inside the mail client and pivoting the indicators across public repositories was straightforward and successful on the first attempt.

5. Conclusion

Completing this room has radically changed my approach to incident triage, shifting my perspective from manually guessing a threat's severity to leveraging structured, open-source intelligence platforms to verify it. I can now confidently intercept raw .eml email files, parse complex transport headers to locate originating IPs, safely defang malicious indicators, and pivot across centralized databases like VirusTotal, Abuse.ch, and Cisco Talos to track advanced adversary signatures. What I learned was how much crucial context can be extracted from tiny artifacts, such as a specific JA3 firewall fingerprint or an embedded Excel macro hash, without ever needing to run or execute the malware itself. In a real-world Security Operations Center (SOC), this connects directly to the daily work of a security analyst, who must quickly separate harmless false alarms from high-severity corporate breaches to protect the network. Moving forward, I intend to proactively build automated lookup playbooks using these exact APIs to dramatically slash my diagnostic timeline during live alert investigations.